# Matheus Hoffmann

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Dear Directors and Research Committee,

I am writing to present myself for research and engineering work at an institute devoted to the ocean, where the observation of the sea, the sharing of ocean data and the forecasting of what the sea will do next are structuring work, and where the results are meant to reach the people and the public policies of the coast. My curriculum vitae is attached; this letter says what I would bring.

Since 2017 I have built forecasting systems that run in production every day and the data and MLOps platforms that keep them running: churn forecasting for a payment company, loss forecasting for an e-commerce fraud platform, audience forecasting for a broadcaster. More recently I turned that practice to the sea, in three works that each have a DOI, a public repository and a reproducible toolchain.

- **Wave modelling on GPUs.** A port of the expensive kernels of the WAVEWATCH III spectral wave model to GPUs that does not change the answer: the nonlinear interaction term is bit-identical to the Fortran on CPU and CUDA backends, and every benchmark carries the command that reproduces it. This is the discipline an operational model needs when it moves to new hardware.
- **An open platform on public data.** marola gathers the sea, wind, wave and tide forecasts and the official bathing-water bulletins of three state agencies for the beaches of Florianópolis, Rio de Janeiro and Salvador, standardises them, and publishes a daily signal with its sources; plain rules decide anything that concerns a swimmer's safety. It is a small, working example of data infrastructure that serves the public directly.
- **A benchmark proposal.** marola-bench proposes a public benchmark for forecasts on the Brazilian coast in the WeatherBench tradition, with ocean-wave scores and in-situ instruments as the ground truth, so that forecasters can be compared at the same observation points over time.

Before industry, I spent two years in an ocean instrumentation laboratory at UFRJ, on signal processing for a Navy underwater-communication programme and on real-time GPS tracking of boats, so I know what the data of a sensor looks like before it reaches a model. I am also an open-water swimmer and a rescue diver: the sea I write software about is one I swim in.

What I would bring to a research group, then, is the engineering between an observation and a forecast that people can use: numerical modelling on GPUs, operational pipelines with verification built in, data platforms that gather and standardise observations from many sources, and the habit of writing the design down, testing it, and publishing the result in the open with a DOI. I would be glad to discuss where this fits the institute's projects, and I am available for a conversation at your convenience.

Yours faithfully,

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The works, each with a DOI: marola 10.5281/zenodo.23224155; ww3-gpu 10.5281/zenodo.23221351; marola-bench 10.5281/zenodo.23352633; closed source 10.5281/zenodo.23333656; picasso 10.5281/zenodo.23311208; nashkell 10.5281/zenodo.23356739.